# msvtx_caloClusterSumE

MC vs data · overview · rank 26 / 98



rank 26 / 98 by endcap D_equal

used by: NN1 barrel, NN1 endcap

rank-identical columns merged into this one:
    summedCaloClusters/clusterSumE
    MS1Vtx_clusE_raw
    MS1Vtx_barrel_clusE
    MS1Vtx_endcap_clusE

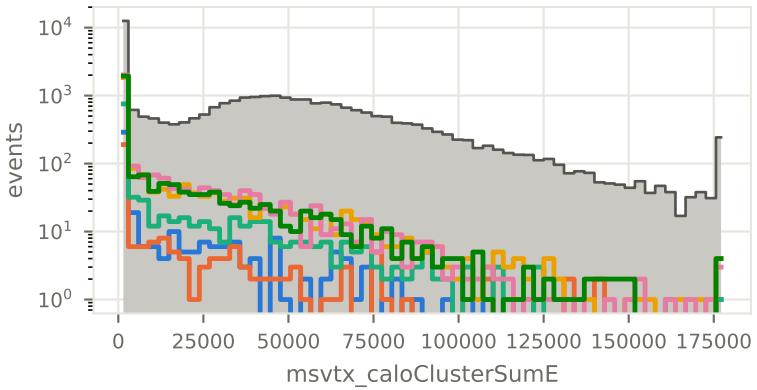
| region | D_equal | noise | D_pooled | 125/15 | 125/55 | 200/50 | 400/100 | 600/150 | 1000/275 | n_bkg | rank |
|---|---|---|---|---|---|---|---|---|---|---|---|
| endcap | 0.383 | 0.022 | 0.386 | 0.392 | 0.372 | 0.381 | 0.379 | 0.389 | 0.396 | 34,970 | 26/98 |
| barrel | 0.310 | 0.051 | 0.312 | 0.343 | 0.282 | 0.330 | 0.327 | 0.318 | 0.292 | 1,089 | 32/98 |
| all | 0.414 | 0.018 | 0.415 | 0.415 | 0.414 | 0.414 | 0.411 | 0.420 | 0.416 | 36,059 | 18/98 |

# msvtx_caloClusterSumE

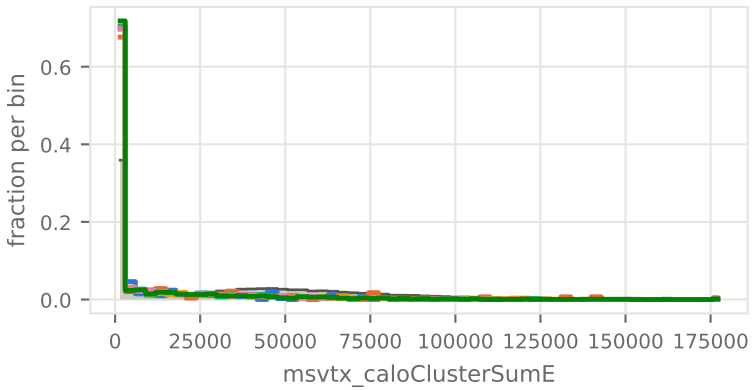
region: ENDCAP · rank 26 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.383 (noise 0.022)  D_pooled 0.386  min 0.372 (mH125_mS55)  max 0.396 (mH1000_mS275)  bkg n = 34,970  sig n = 9,996

1a · each mass point vs background · raw

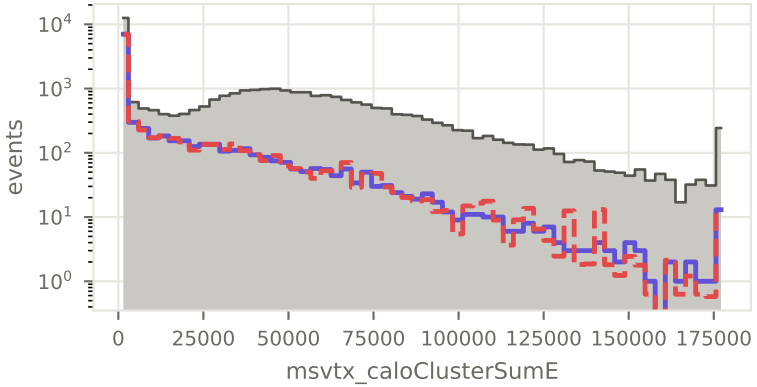



1b · each mass point vs background · normalized

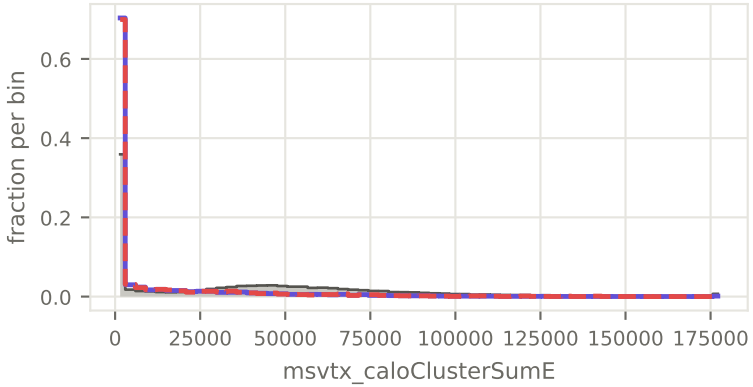



1 · each mass point
background n 34,970
mH125_mS15 D 0.392 n 411
mH125_mS55 D 0.372 n 281
mH200_mS50 D 0.381 n 1,071
mH400_mS100 D 0.379 n 2,648
mH600_mS150 D 0.389 n 2,882
mH1000_mS275 D 0.396 n 2,703

2a · combined signal vs background · raw

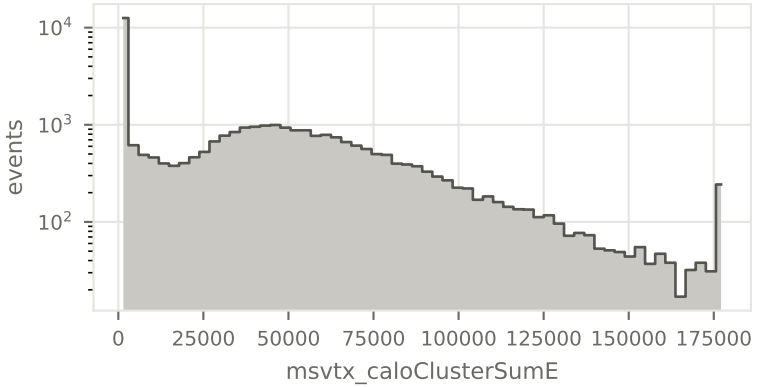



2b · combined signal vs background · normalized

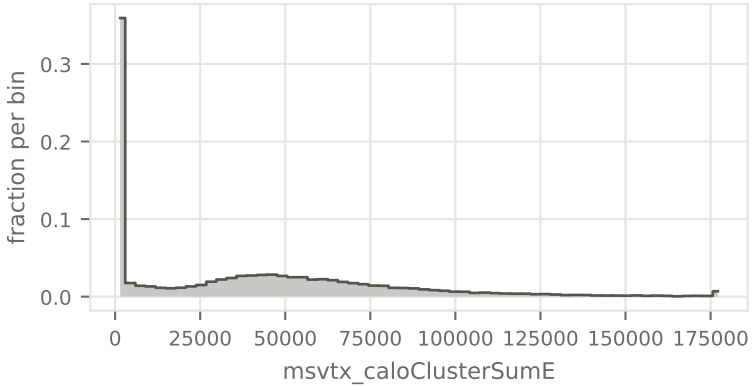



2 · combined signal
background n 34,970
combined, pooled D 0.386 n 9,996
equal weight per mass D 0.383

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

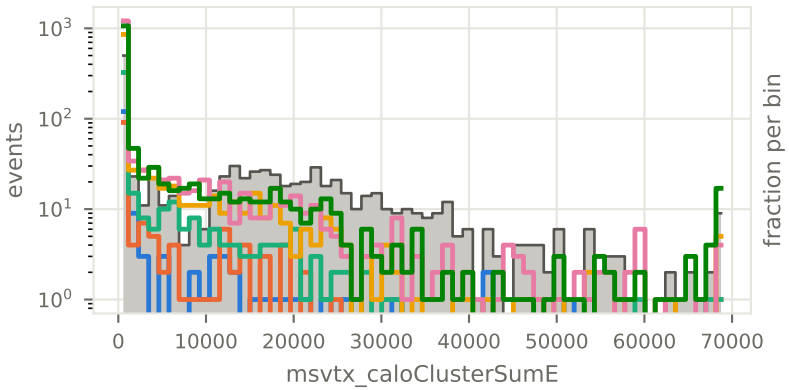
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_caloClusterSumE

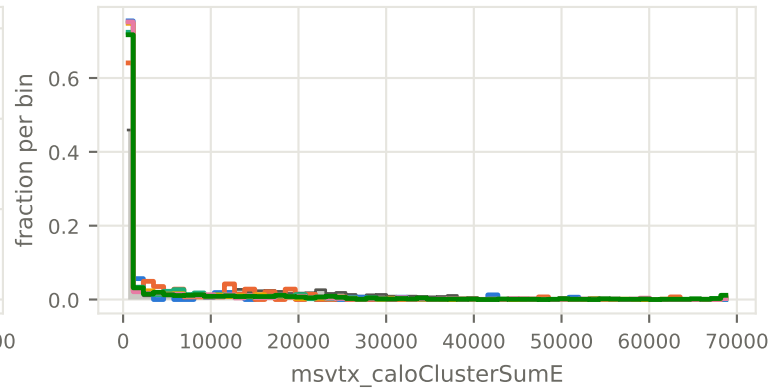
region: BARREL · rank 26 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.310 (noise 0.051) D_pooled 0.312 min 0.282 (mH125_mS55) max 0.343 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

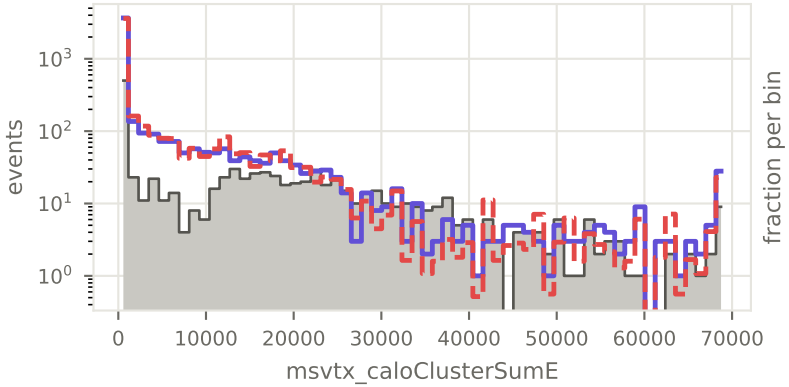



1b · each mass point vs background · normalized

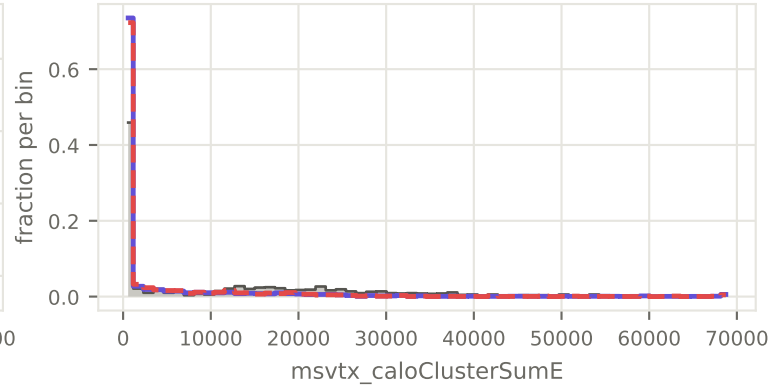



1 · each mass point
background n 1,089
mH125_mS15 D 0.343 n 159
mH125_mS55 D 0.282 n 142
mH200_mS50 D 0.330 n 449
mH400_mS100 D 0.327 n 1,144
mH600_mS150 D 0.318 n 1,608
mH1000_mS275 D 0.292 n 1,485

2a · combined signal vs background · raw

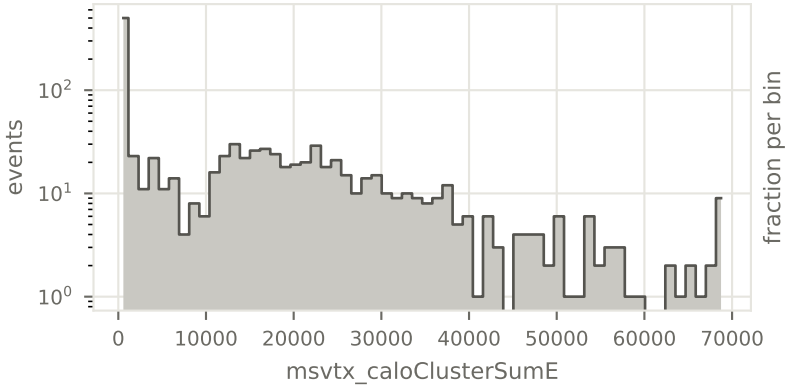



2b · combined signal vs background · normalized

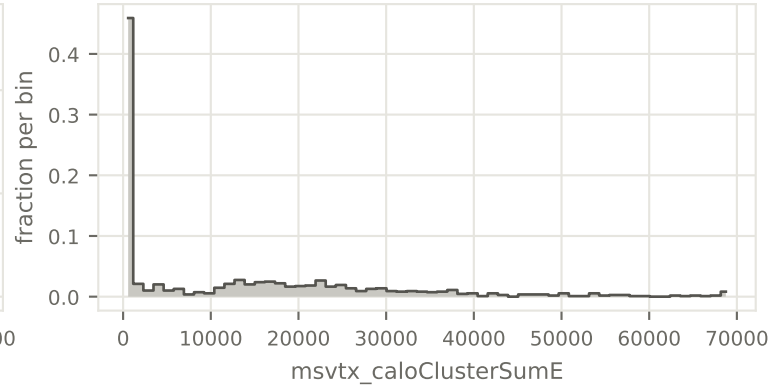



2 · combined signal
background n 1,089
combined, pooled D 0.312 n 4,987
equal weight per mass D 0.310

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

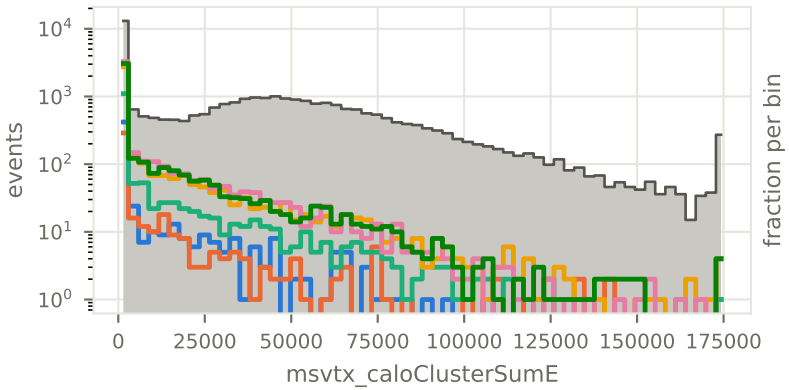
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_caloClusterSumE

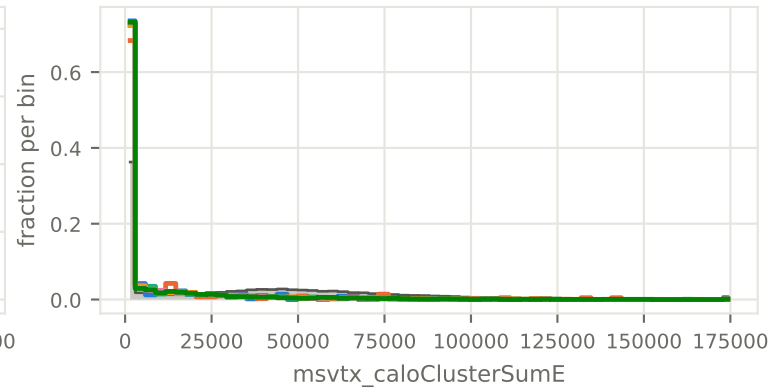
region: ALL · rank 26 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.414 (noise 0.018)  D_pooled 0.415  min 0.411 (mH400_mS100)  max 0.420 (mH600_mS150)  bkg n = 36,059  sig n = 14,983

1a · each mass point vs background · raw

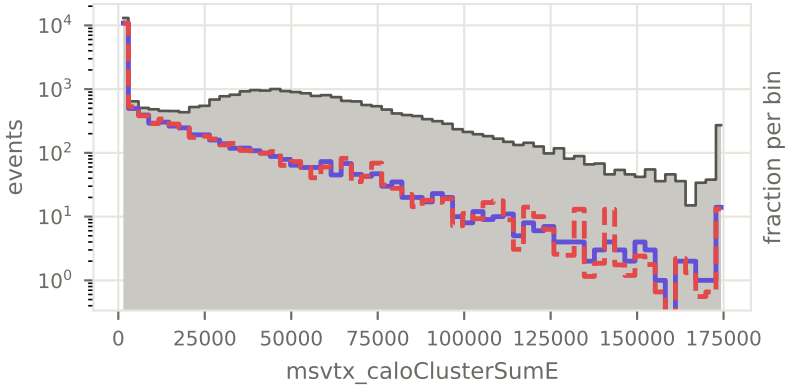



1b · each mass point vs background · normalized

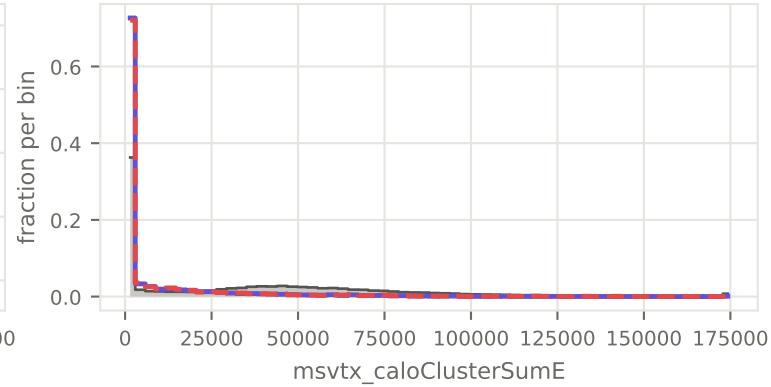



1 · each mass point
background n 36,059
mH125_mS15 D 0.415 n 570
mH125_mS55 D 0.414 n 423
mH200_mS50 D 0.414 n 1,520
mH400_mS100 D 0.411 n 3,792
mH600_mS150 D 0.420 n 4,490
mH1000_mS275 D 0.416 n 4,188

2a · combined signal vs background · raw

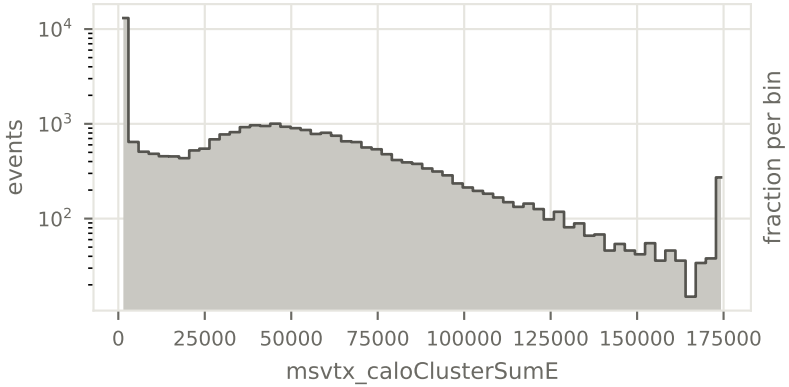



2b · combined signal vs background · normalized

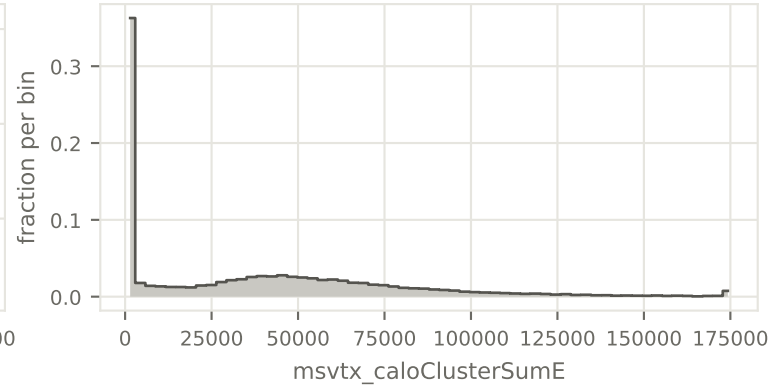



2 · combined signal
background n 36,059
combined, pooled D 0.415 n 14,983
equal weight per mass D 0.414

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed